PennState College of Engineering

HAROLD AND INGE MARCUS DEPARTMENT OF INDUSTRIAL AND MANUFACTURING ENGINEERING

Research

NeurMHSP

Taehyeon Kwon

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Overview

Overview

- Interpret equations, estimating number of strategic and operational node (done)
- Read a 'EMPIRE' paper and implementation document (ongoing)

Today work

Today work

- Interpret following table (kaut et al., 2014.)

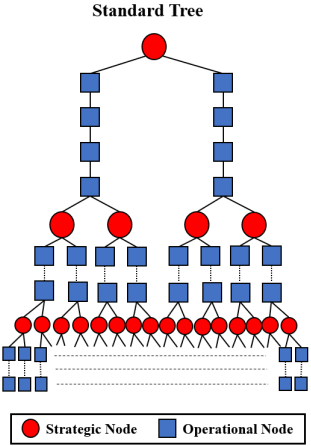
Table 1 Numbers of strategic and operational nodes for the standard tree in Fig. 2c and the multi-horizon tree in Fig. 3b

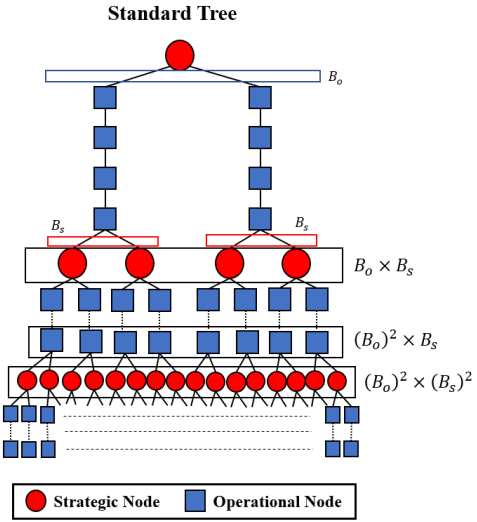
	Standard tree	Multi-horizon tree
# strategic nodes	$\frac{1-(B_o B_s)^{P_s}}{1-B_o B_s} \approx (B_o B_s)^{P_s}$	$\frac{1-(B_s)^{P_s}}{1-B_s} \approx B_s^{P_s}$
# operational nodes	$P_o B_o \frac{1-(B_o B_s)^{P_s}}{1-B_o B_s} \approx P_o B_o (B_o B_s)^{P_s}$	$P_o B_o \frac{1-(B_s)^{P_s}}{1-B_s} \approx P_o B_o B_s^{P_s}$

P_s is the number of strategic periods, P_o the number of operational periods per strategic period. B_s and B_o are the numbers of strategic and operational successors in each strategic and operational branching, respectively. We assume that these numbers are constant throughout the tree and that $B_s > 1$

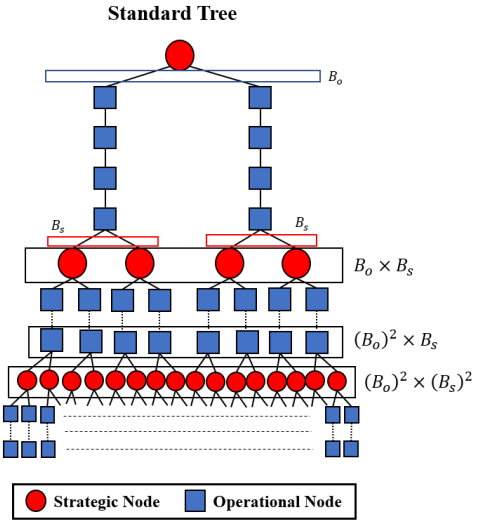
Standard Tree

- B_s should be larger than 1. In other words, $B_s \geq 2$ for integer values.
- Let's consider the following standard tree.
  - $P_s = 3, P_o = 4, B_s = 2, B_o = 2$





- P_s should also be greater than or equal to two, because $B_s \geq 2$. If P_s is less than 2, we cannot satisfy the condition $B_S > 1$.
- Basically, to estimate the number of strategic and operational nodes, we must have at least two-stage trees.



- Based on the following figure, we can easily understand the pattern of the number of strategic nodes. Based on these patterns, we can define the number of strategic nodes as follows:

$$\begin{aligned} &\#\text{ of strategic nodes} \\ &= 1 + (B_o B_s)^1 + (B_o B_s)^2 \end{aligned}$$

- If, $P_s = n, n \geq 2$,

$$\begin{aligned}\#\text{ of strategic nodes} &= 1 + (B_oB_s)^1 + (B_oB_s)^2 + \cdots + (B_oB_s)^{n-1} \\ &= \sum_{a=1}^{n} (B_oB_s)^{a-1}\end{aligned}$$

- More specifically,

$$\begin{aligned}\sum_{a=1}^{P_s} (B_oB_s)^{a-1} &= \frac{(B_oB_s)^0((B_oB_s)^{P_s} - 1)}{B_oB_s - 1} \\ &= \frac{(B_oB_s)^{P_s} - 1}{B_oB_s - 1}\end{aligned}$$

- The same logic can be applied to estimate the number of operational nodes in the standard tree.
- Operational nodes are highly impacted by B_o and P_o. $B_o \times P_o$ indicates the number of operational nodes derived from one strategic node.
- We can estimate the number of operational nodes in the standard tree as follows:

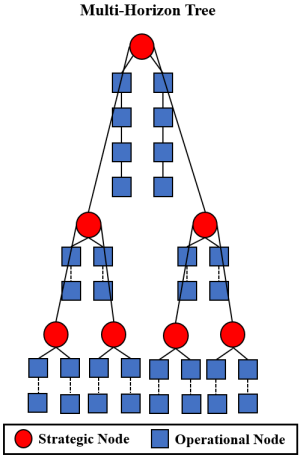
$$
\begin{aligned}
\text{\# of operational nodes} &= B_oP_o \times (\text{\# of strategic node at stage 1}) \\
&+ B_oP_o \times (\text{\# of strategic node at stage 2}) \\
&+ B_oP_o \times (\text{\# of strategic node at stage 3}) \\
&+ \cdots \\
&= \sum_{n=1}^{P_s} B_oP_o \times (\text{\# of strategic node at stage n}) \\
&= \sum_{n=1}^{P_s} B_oP_o \times (B_oB_s)^{n-1}
\end{aligned}
$$

- More specifically,

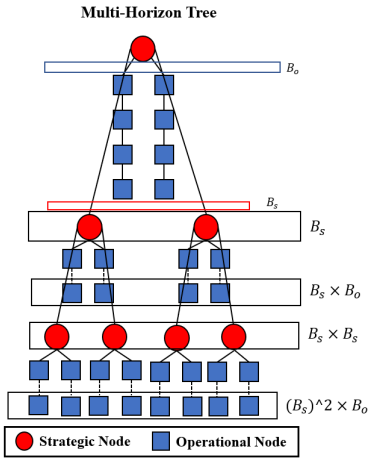
$$\begin{aligned}\sum_{n=1}^{P_s} B_o P_o \times (B_o B_s)^{n-1} &= B_o P_o \sum_{n=1}^{P_s} (B_o B_s)^{n-1} \\ &= B_o P_o \times \left(\frac{(B_o B_s)^{P_s} - 1}{B_o B_s - 1}\right) \\ &= B_o P_o \frac{(B_o B_s)^{P_s} - 1}{B_o B_s - 1}\end{aligned}$$

Multi-Horizon Tree

- Let's move on to Multi-horizon tree.
- Under the same conditions as the standard tree: $P_s = 3, P_o = 4, B_s = 2, B_s = 2$.



- The following figure helps us understand the number of strategic nodes in a multi-horizon setting.



- Unlike the standard tree, the number of strategic nodes is not affected by the number of options for the operational decision, B_o.
- Thus, the number of strategic nodes changes based on B_s.

$$\#\ \text{of strategic nodes} = 1 + B_s + B_s^2$$

- If, $P_s = n, n \geq 2$,

$$\begin{aligned} \#\text{ of strategic nodes} &= 1 + (B_s)^1 + (B_s)^2 + \cdots + (B_s)^{n-1} \\ &= \sum_{a=1}^{n} (B_s)^{a-1} \end{aligned}$$

- More specifically,

$$\begin{aligned} \sum_{a=1}^{P_s} (B_s)^{a-1} &= \frac{(B_s)^0((B_s)^{P_s} - 1)}{B_s - 1} \\ &= \frac{(B_s)^{P_s} - 1}{B_s - 1} \end{aligned}$$

- Likewise, the same logic can be applied to estimate the number of operational nodes in the multi-horizon tree.
- We can estimate the number of operational nodes in the multi-horizon tree as follows:

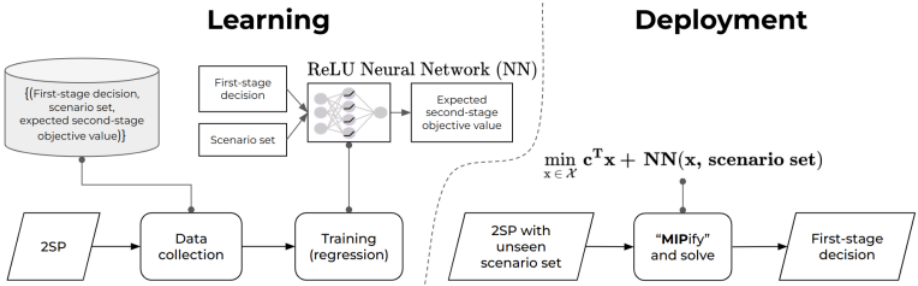
$$
\begin{aligned}
\text{\# of operational nodes} &= B_o P_o \times (\text{\# of strategic node at stage 1}) \\
&+ B_o P_o \times (\text{\# of strategic node at stage 2}) \\
&+ B_o P_o \times (\text{\# of strategic node at stage 3}) \\
&+ \cdots \\
&= \sum_{n=1}^{P_s} B_o P_o \times (\text{\# of strategic node at stage n}) \\
&= \sum_{n=1}^{P_s} B_o P_o \times (B_s)^{n-1}
\end{aligned}
$$

- More specifically,

$$\begin{aligned}\sum_{n=1}^{P_s} B_o P_o \times (B_s)^{n-1} &= B_o P_o \sum_{n=1}^{P_s} (B_s)^{n-1} \\ &= B_o P_o \times \left(\frac{(B_s)^{P_s} - 1}{B_s - 1}\right) \\ &= B_o P_o \frac{(B_s)^{P_s} - 1}{B_s - 1}\end{aligned}$$

Neur2SP

- How can we generate data for training Neural Network (NN).



- What dose mean ’aggregated scenarios’?

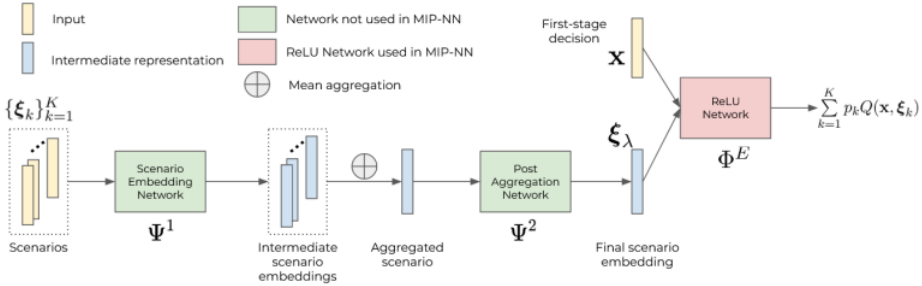


Figure 2: NN-E architecture diagram.

Future Work

Future works

- Study the stochastic programming with a book 'Introduction to Stochastic Programming' (continuously)
- Fully understand the Neur2SP model.
- Complete to read a EMPIRE paper and understand the concepts.
- Implement the EMPIRE software.
- Understand the method of scenario generation.

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